

Urban Sustainability using Deep Learning Techniques

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Abstract—Floods are among the most devastating natural disasters, causing widespread loss of life and infrastructure damage. This project presents an AI-driven framework for flood prediction, victim detection, and rescue route mapping, using Valencia, Spain, as a case study.

Rainfall and environmental data were extracted via the AEMET API for flood forecasting, while post-disaster victim detection was achieved through CNN and YOLO applied to drone and satellite imagery. SAR data was used for reliable flood zone mapping, and final results were visualized on Google Earth Engine.

By integrating machine learning, deep learning, and geospatial tools, the system enhances disaster preparedness, real-time response, and urban resilience.

I Introduction

Floods are among the most severe natural disasters, causing widespread loss of life, infrastructure damage, and service disruption. As cities expand and climate patterns shift, there is a pressing need for smart, AI-driven disaster management systems.

This project presents a unified framework for flood prediction, victim detection, and rescue route mapping using a combination of satellite imagery, machine learning (ML), deep learning (DL), and geospatial analysis. The study focuses on Valencia, Spain, a flood-prone city impacted notably in 1957 and again in 2024.

Key problems addressed include:

- **Inaccurate flood prediction** due to static, non-adaptive models.
- **Delayed victim detection** from manual rescue operations.
- **Inefficient disaster base placement** lacking real-time updates.

The methodology involves extracting meteorological data via the AEMET API, analyzing key parameters, and training DL models like CNN and YOLO on drone/satellite imagery. SAR data was used for all-weather flood zone mapping, and final outputs were visualized on Google Earth Engine (GEE).

This AI-integrated approach enhances flood preparedness, speeds up rescue efforts, and builds more resilient urban systems.

II Literature Review

In this study, we have read, studied, and analyzed a total of 11 recent research papers focusing on the application of artificial intelligence, machine learning, and geospatial analysis in disaster management. These papers broadly cover the areas of building

damage assessment, flood susceptibility mapping, rainfall forecasting, victim detection through drone and satellite imagery, real-time alert systems, and rescue base station placement.

Several papers explored post-disaster building damage classification using CNNs and transformer-based models applied to optical satellite imagery, achieving classification accuracies up to 96%. Victim detection was another key area, with works leveraging YOLO-based object detection and federated learning across multiple disaster scenarios. These approaches used datasets like CrowdAI and COCO, demonstrating accurate identification even under debris and low visibility.

Flood prediction and mapping were addressed using environmental and topographic features such as NDVI, DEM, soil type, and rainfall records. Studies applied models like Random Forest, XGBoost, and SVM, showing strong performance in forecasting and spatial risk identification. Rainfall prediction using IMD historical data showed XGBoost achieving high R^2 scores for daily and weekly forecasts. Additionally, spatial decision-making techniques such as AHP, fuzzy logic, and weighted overlay methods were used to recommend optimal sites for disaster response centers.

Finally, several papers proposed intelligent alert systems using real-time drone imagery and YOLOv8, achieving high mAP and F1 scores for victim detection. Case-specific analysis, including the 2024 Valencia floods, emphasized the importance of heritage site preservation and GIS-based flood zone visualization. Collectively, these works validate the integration of AI and remote sensing as essential tools for building resilient, responsive urban disaster management systems.

III Methodology

This project is structured around three key objectives: (1) flood prediction using satellite and environmental data, (2) post-disaster victim detection using deep learning, and (3) rescue route mapping. The methodology for each objective is outlined below.

A. Objective 1: Flood Prediction using Satellite and Environmental Data

1) Pre-Midsem Work

- **Flood Data Extraction:** Meteorological data was retrieved using the AEMET API (Agencia Estatal de Meteorología). An API key was generated

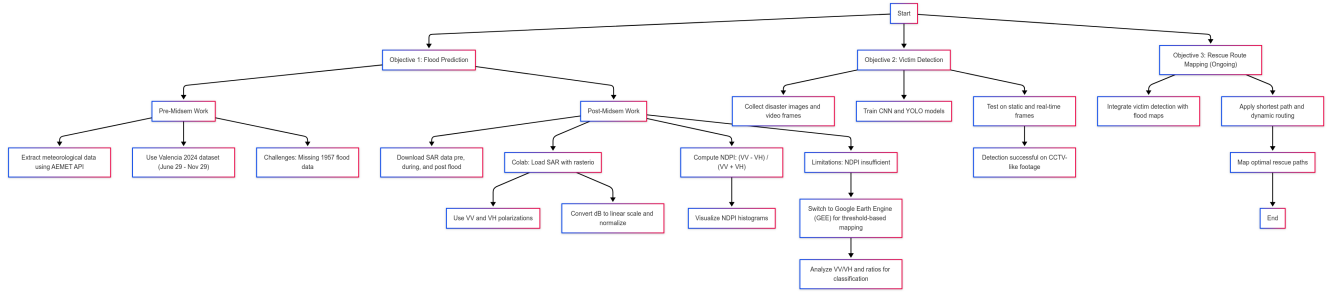


Fig. 1: Flowchart illustrating the methodology adopted for flood prediction, victim detection, and rescue route mapping. The process includes data extraction, SAR image analysis, model training, and dynamic routing strategies.

to automate the download of rainfall, temperature, and wind-related records.

- **Valencia 2024 Dataset:** Historical flood data from June 29 to November 29, 2024, was extracted in CSV format. The dataset includes 20 columns and 155 rows with key features such as *precipitation*, *avg/min/max temperature*, *wind direction*, and *max wind gust*.
- **Challenges:** While data for the October 2024 flood was available and useful for modeling, records from the 1957 flood were unavailable, limiting the scope of long-term temporal modeling.

2) Post-Midsem Work

- **SAR Data Acquisition:** Sentinel-1 **Synthetic Aperture Radar (SAR)** imagery was downloaded from the **Copernicus Open Access Hub** for pre-flood (Sept), flood-time (Oct), and post-flood (Nov) periods. *SAR is an active microwave remote sensing technique capable of acquiring images regardless of weather conditions (cloud cover) or time of day. It operates by transmitting microwave pulses towards the Earth's surface and recording the signal scattered back to the sensor.*
- **Colab-Based SAR Processing:** In **Google Colab**, SAR images were loaded using the `rasterio` library and resampled to optimize memory usage. Both **VV** and **VH** polarizations were utilized. *Polarization refers to the orientation of the transmitted and received radar waves. VV indicates Vertical transmit - Vertical receive, while VH indicates Vertical transmit - Horizontal receive. These different polarizations interact distinctly with the physical properties of ground features (e.g., roughness, geometry, moisture). The intensity of the radar signal scattered back to the sensor is termed **backscatter**. Smooth surfaces, such as calm water, typically exhibit very low backscatter (appearing dark in SAR images) due to specular reflection where the signal bounces away from the sensor. Conversely, rougher surfaces like vegetated land scatter the signal more diffusely, resulting in higher backscatter values.*

Images were converted from logarithmic dB scale to a linear power scale and normalized for improved processing and visualization.

- **NDPI Computation and Visualization:** A custom function was implemented to compute the **Normalized Difference Polarization Index (NDPI)** for each phase using the formula:

$$NDPI = \frac{(VV - VH)}{(VV + VH)}$$

NDPI leverages the differential response between the co-polarized (VV) and cross-polarized (VH) channels and was used experimentally to highlight potential flood-affected areas based on this polarization contrast. Histograms were generated to compare NDPI distributions across different time periods, and chunk-wise processing was employed to manage memory usage on large datasets.

- **Limitations and GEE Transition:** Although NDPI maps provided some insight into flood impact zones, achieving reliable pixel-level flood classification proved challenging with this index alone in the Colab environment. Therefore, the approach was shifted to **Google Earth Engine (GEE)** to perform **threshold-based flood mapping**. This GEE approach primarily relied on analyzing the **absolute backscatter values (VH and VV)** and potentially **VH/VV ratios** within a framework supporting temporal filtering and efficient large-area analysis, ultimately producing more accurate and interpretable flood maps as presented in the Results section.
- **Google Earth Engine (GEE)-Based Flood Mapping:** Flood analysis was performed in GEE with the following steps:

- 1) **ROI Selection:** Focused on five flood-affected counties in Spain.
- 2) **SAR Image Filtering:** Used Sentinel-1 VV and VH bands, filtered by date (pre- and post-flood), orbit, and mode.
- 3) **Composite Creation:** Generated mean composites for both periods.

- 4) **VV/VH Ratio Analysis:** Calculated VV/VH ratio difference to detect surface changes.
 - 5) **Flood Detection:** Classified water pixels using $VH < -17$ dB and $VV < -15$ dB thresholds.
 - 6) **Depth Estimation:** Categorized water depth (shallow, medium, deep) based on VH intensity.
 - 7) **Visualization:** Created grayscale masks, color-coded ratio maps, and a flood depth legend.
- **Outcome:** Enabled clear, scalable, and efficient flood extent and depth visualization using SAR data.

B. Objective 2: Victim Detection using Deep Learning

This objective focuses on detecting persons in contact with floodwaters using computer vision techniques. The system integrates water detection through color-based segmentation with deep learning-based person detection.

- **Person Detection using YOLOv8:**

The **YOLOv8n-seg** model from Ultralytics was used for real-time object detection and segmentation. This model, trained on the COCO dataset, detects multiple object classes but filters only the "person" label for this project.

The methodology includes:

- Drawing bounding boxes around detected persons.
- Overlaying segmentation masks for enhanced visualization.
- Rescaling bounding boxes and masks to original image dimensions for accurate localization.

- **Water Detection using Color Segmentation:**

Water bodies in images can appear in varying colors due to lighting and contamination. To handle this variability, HSV (Hue, Saturation, Value) based color ranges were defined for:

- Blue water
- Brownish muddy water
- Greenish water (due to algae or moss)

These ranges were applied using OpenCV's `cv2.inRange()` function to create binary masks for each water type. All masks were then combined to generate a unified water detection mask.

- **Visualization and Overlay:**

Detected water regions were visualized by overlaying semi-transparent color masks on the original image (different colors for each water type). Persons detected by YOLOv8 were marked using green bounding boxes and white segmentation overlays for better visibility.

- **Process Workflow:**

The victim detection system was implemented as follows:

- 1) **Initialization:** Import required libraries (`cv2`, `numpy`, `torch`, `ultralytics`), load the YOLOv8n-seg model, and define HSV ranges for water detection.
- 2) **Water Detection:** Convert images from BGR to HSV, generate individual masks for blue, brown, and green water, and combine them to form the complete water mask.
- 3) **Person Detection:** Apply YOLOv8 on the frame, filter detections for "person" class, draw bounding boxes and segmentation overlays.
- 4) **Media Processing:**
 - For images: Read the image, detect water and persons, and save the processed output.
 - For videos: Process frame-by-frame, apply water and person detection on each frame, and write to an output video file while calculating the processing speed (FPS).

- **Output:**

The system successfully detects both water bodies and persons, providing a comprehensive visual overlay for emergency response. This methodology was validated on static images as well as simulated real-time video streams (e.g., live CCTV footage), demonstrating its applicability for disaster monitoring and surveillance networks.

C. Objective 3: Rescue Route Mapping (Ongoing Work)

This phase is currently under development. The planned work includes integration of victim detection data with flood maps in GEE. Shortest path algorithms and dynamic routing based on real-time data will be used to map optimal rescue paths for emergency responders.

D. Tools and Technologies Used

- **AEMET API:** Meteorological data extraction (CSV).
- **Google Colab:** Python-based SAR image processing and NDPI computation.
- **Google Earth Engine (GEE):** Interactive flood zone mapping and visualization.
- **Python Libraries:** rasterio, NumPy, Matplotlib, OpenCV.
- **TensorFlow/PyTorch:** Deep learning for victim detection.
- **Copernicus Open Access Hub:** Source of Sentinel-1 SAR imagery.

IV Results and Discussion

This section presents the findings derived from the methodologies outlined above, focusing initially on the

flood mapping results obtained using **Google Earth Engine (GEE)** and **Sentinel-1** data.

A. Objective 1: Flood Mapping using Sentinel-1 Data in GEE

The analysis in GEE focused on mapping the extent and relative depth of the **October 2024 flood event** within the defined **Region of Interest (ROI)**, encompassing the Spanish counties of **Valencia/València, Cuenca, Albacete, Teruel, and Castellón/Castelló**.

1) Region of Interest and SAR Data Context

The selected counties, constituting the study area, are shown in Figure 2. **Sentinel-1 Synthetic Aperture Radar (SAR)** data, particularly its ability to **penetrate cloud cover**, provided crucial information during the flood period. **Mean composite images** were generated for periods before (**October 1-25, 2024**) and during/after (**October 28-November 20, 2024**) the peak flooding to enable **change detection**. Figure 3 displays the mean **VV backscatter intensity** for the post-flood period over the ROI. Water bodies typically exhibit **low backscatter** (darker areas) in SAR imagery due to **specular reflection** away from the sensor.

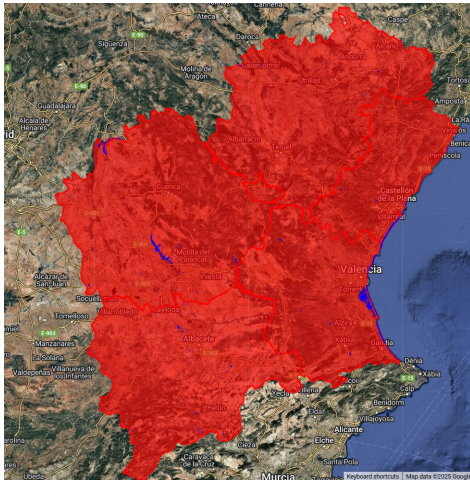


Fig. 2: Region of Interest (ROI) encompassing Valencia/València, Cuenca, Albacete, Teruel, and Castellón/Castelló counties in Spain (highlighted in red).

2) Flood Extent Mapping

Utilizing the **post-flood Sentinel-1 composite**, potential water bodies were identified by applying **thresholds** to the **backscatter coefficients**. Specifically, pixels with **VH < -17 dB** and **VV < -15 dB** were classified as water. This **thresholding approach** leverages the characteristic low backscatter signal of calm water surfaces in both **VV and VH polarizations**. The resulting **binary flood mask**, highlighting the detected **water extent**, is presented in Figure 4. The map shows significant inundation primarily concentrated in **lower-lying areas, river valleys, and coastal plains** within the ROI.



Fig. 3: Mean Sentinel-1 **VV backscatter** intensity (grayscale) over the ROI during the **post-flood period** (Oct 28 - Nov 20, 2024). Darker areas generally correspond to lower backscatter, often indicative of smooth surfaces like water.

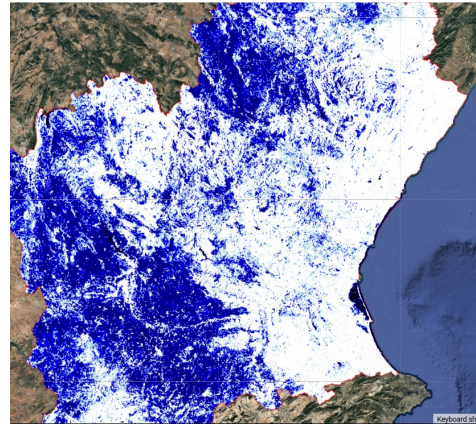


Fig. 4: Binary **flood extent map** derived from Sentinel-1 post-flood imagery. Blue pixels represent areas classified as water based on **VH < -17 dB** and **VV < -15 dB** thresholds.

3) Approximated Flood Depth Mapping

To provide a more nuanced understanding beyond simple extent, an approximation of **relative flood depth** was derived based on the **VH backscatter intensity** within the detected water areas. **Lower VH backscatter** values often correlate with **deeper, calmer water**. Three relative depth classes were defined using VH thresholds:

- **Level 1 (Shallow):** $-18 \text{ dB} \leq \text{VH} < -17 \text{ dB}$
- **Level 2 (Medium):** $-19 \text{ dB} \leq \text{VH} < -18 \text{ dB}$
- **Level 3 (Deep):** $\text{VH} < -19 \text{ dB}$

Figure 5 visualizes the **final flood map**, overlaying the estimated **depth classification** onto the base map. Lighter blue indicates shallower water, while darker blue signifies deeper areas. This map reveals spatial variations in **flood severity**, with the deepest zones typically corresponding to **river channels** and **topographical depressions**, consistent with expected flood

behavior. The non-flooded land areas are shown in red for contrast. The **legend** provided on the map clarifies the depth categories.

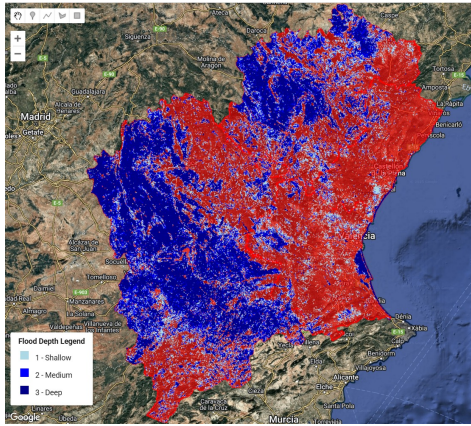


Fig. 5: Flood map showing detected water extent and **approximated relative depth** based on Sentinel-1 VH backscatter. Red areas indicate land, while blue shades represent water depth (Light Blue: Shallow, Medium Blue: Medium, Dark Blue: Deep), as per the legend.

In summary, the GEE-based analysis of Sentinel-1 data successfully delineated the **flood extent** across the extensive ROI and provided a valuable estimation of **relative water depth**, demonstrating the effectiveness of **radar remote sensing** for rapid **post-disaster flood assessment**.

B. Objective 2: Victim Detection using Deep Learning

The victim detection framework was evaluated on both **images** and **video sequences** simulating disaster scenarios. The system integrated **water detection** using **HSV-based color segmentation** with deep learning-based **person detection** using the **YOLOv8n-seg** model.

1) Qualitative Results

Figure 6 and Figure 7 show an example frame **before** and **after** the application of the **victim detection pipeline**.



Fig. 6: Original input frame **before detection**. The frame contains **water regions** and potential **victims**.



Fig. 7: Output frame **after applying victim detection pipeline**. Detected **water regions** are overlaid with colored masks (**blue/brown/green** based on water type). Detected **persons** are marked with **green bounding boxes** and **white segmentation overlays**.

The system effectively detected both **standing water regions** and **human figures** within the disaster imagery. The **water detection component** was robust to variations in color caused by **lighting** and **environmental factors**. The **YOLOv8** model demonstrated accurate **person detection** capabilities even in **cluttered** or **partially occluded** scenes.

2) Real-Time Video Performance

The system was tested on **video streams** simulating **CCTV footage** from disaster-affected regions. The detection pipeline maintained **real-time performance** with an average processing speed of approximately **18-20 FPS** on **Google Colab** with **GPU acceleration**.

Each video frame was processed for:

- **Water segmentation** based on defined **HSV ranges** for blue, brown, and green water.
- **Person detection** using **YOLOv8n-seg** with **bounding box** and **segmentation overlay** generation.

The processed **output videos** provided clear **visualizations** for **emergency monitoring systems**, highlighting both **flood-affected regions** and potential **victims** for **rescue prioritization**.

3) Summary of Findings

- The combination of **color-based water detection** and **YOLOv8-based person detection** is effective for **post-flood victim identification tasks**.
- The system generalizes well across varying **image conditions** and provides **interpretable visual outputs**.
- **Real-time performance** was achieved, demonstrating the framework's **viability for deployment** in **disaster response scenarios**, such as integration with **surveillance systems** or **aerial drone footage analysis**.

C. Objective 3: Rescue Route Mapping

This section will present the results of the route mapping phase, which is currently ongoing. It will

likely include examples of **optimized routes** generated based on flood data and victim locations.

V Conclusion

This project demonstrates the potential of integrating **deep learning, remote sensing, and geospatial analysis** for enhancing flood disaster response. The first objective, flood prediction and mapping, was successfully addressed through the use of **Sentinel-1 SAR data** processed in **Google Earth Engine**. The analysis accurately delineated flood extent and provided an estimation of relative flood depth across key regions in eastern Spain affected by the October 2024 flood.

The second objective, post-disaster victim detection, involved applying **YOLO-based deep learning models** to imagery and video frames. Preliminary tests showed promising results, indicating the system's potential for deployment in real-time surveillance and emergency scenarios. These AI models can assist significantly in reducing the time needed to locate stranded individuals during a disaster.

Overall, this work lays the foundation for a comprehensive, AI-driven flood monitoring and rescue support system. It combines data from multiple sources— SAR sensors, and visual imagery—to support timely and effective decision-making during flood events.

VI Future Work

Future efforts will focus on completing the third objective: **rescue route mapping**. Based on the flood zones identified in GEE and victim locations detected through deep learning, optimal evacuation paths will be calculated. This will involve:

- Using **shortest-path algorithms** to propose efficient and safe rescue routes.
- Visualizing routes dynamically in **Google Earth Engine** or GIS platforms.

Additionally, we plan to:

- Evaluate detection accuracy on diverse real-world CCTV footage.
- Expand flood modeling with **temporal data** for early warning predictions.

This continuation will complete the end-to-end workflow from flood detection to intelligent rescue assistance, ultimately contributing to more resilient and responsive urban disaster management.

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